University of Florida, Department of Mathematics

Topology PhD Exam, May 2014

Work the following problems and show all work. Support all statements to the best of your ability.

1. Compute $\pi_1(T^2)$ and $H_*(T^2)$, where $T^2 = S^1 \times S^1$ is the 2-dimensional torus.

2. Let X be a connected completely regular topological space having more than one point. Can X be countable?

3. Prove that there is no map of degree 3 from S^2 to the torus T^2.

4. Does there exist a covering space of the 2-dimensional sphere with four points deleted with nontrivial abelian fundamental group?

5. Show that the 2-dimensional sphere with four points deleted cannot be a topological group.

Answer the following with complete definitions or statements or short proofs.

6. State the Baire Category Theorem.

7. State the homology Mayer–Vietoris Theorem.

8. State the Urysohn Lemma.

9. Do the following short exact sequences always split

$$0 \to A \to B \to \mathbb{Z}^2 \to 0$$

$$0 \to \mathbb{Z} \to A \to B \to 0\ ?$$

10. Compute the Euler characteristic $\chi(\mathbb{R}P^2 \times S^1 \times S^2 \times S^3)$.

11. Give an example of $A \subset X$ such that A is a retract of X but not a deformation retract.

12. State the Lefschetz Fixed Point Theorem.

13. Can irrational numbers be presented as a countable union of closed in $\mathbb{R}$ subsets?

14. Draw a picture of the universal cover of the 2-sphere with the segment joining the north and south poles.

15. List all i for which there is a closed orientable 6-manifold M with $H_i(M) = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$.